

2024 Greenhouse Gas Emission Inventory for the Royal Belgian Institute for Space Aeronomy (BIRA-IASB)



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Context

In line with its commitment to limiting the environmental impact of its activities, the Royal Belgian Institute for Space Aeronomy (BIRA-IASB) has developed a greenhouse gas (GHG) inventory and estimated the institute's carbon footprint for the year 2024, with the following results:

Total estimated carbon footprint

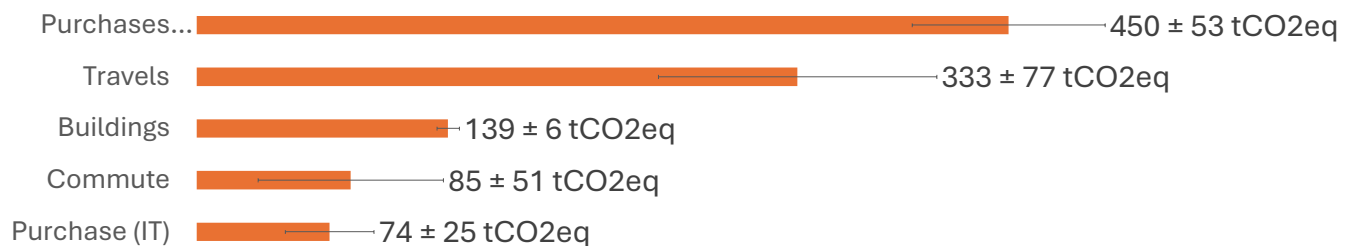
$1081 \pm 82 \text{ tCO}_2\text{eq}$

Carbon intensity per € spent

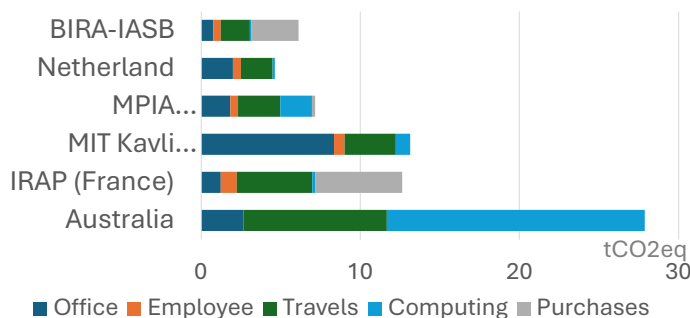
$57 \pm 4 \text{ gCO}_2\text{eq/€}$

Carbon footprint per capita

$5.7 \pm 0.4 \text{ tCO}_2\text{eq}$



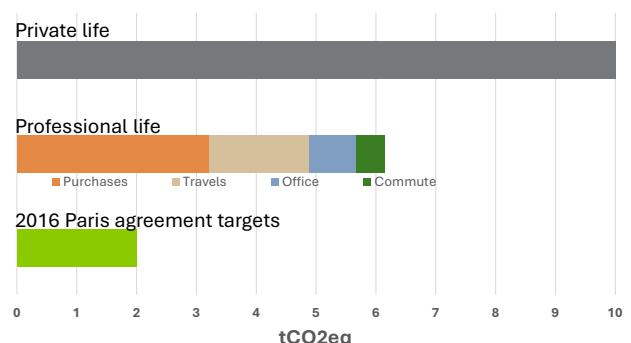
BIRA-IASB carbon footprint compared to other institutes



Based on ‘Section 18.3 The Carbon Footprint of Astronomical Research (Knödseder, Jürgen)’, in *Climate Change for Astronomers: Causes, consequences, and communication*, IOP Astronomy, 2024.

Conclusions

In Europe, the average annual carbon footprint per person for private activities is approximately 10 tCO₂e. At BIRA-IASB, scientists generate an additional 6 tCO₂e through their professional activities, resulting in a total footprint of about 16 tCO₂e per person—roughly eight times higher than the Paris Agreement target of 2 tCO₂e needed to limit further global warming. Achieving substantial reductions will require a paradigm shift in our field, including rethinking how scientific work is organised, conducted, and valued.



Tools

GES 1 POINT 5